

Jail Database – CSC4710

The city's Crime Analysis Unit has submitted the following data requests. Provide the SQL statements using subqueries to satisfy the requests. Test the statements and show the execution results.

1. List the name of each officer who has reported more than the average number of crimes officers have reported.
2. List the criminal names of all criminals who have a less-than-average number of crimes and aren't listed as violent offenders.
3. List appeal information for each appeal that has a less-than-average number of days between the filing and hearing dates.
4. List the names of probation officers who have had a less-than-average number of criminals assigned.
5. List each crime with the highest number of appeals recorded.

The city's Crime Analysis unit has submitted the following data requests. Provide the SQL statements using subqueries to satisfy the requests. Test the statements and show execution results.

Use a SQL sub-query statement to answer the following: **(Please provide the correct answer for questions 2 to 5)**

6. List the name of each officer who has reported more than the average number of crimes officers have reported.

CORRECT ANSWER:

```
SELECT co.officer_id, o.last, o.first
FROM crime_officers co JOIN officers o
ON co.officer_id = o.officer_id
GROUP BY co.officer_id, o.last, o.first
HAVING COUNT(*) > (SELECT COUNT(*) / COUNT(DISTINCT officer_id) FROM
crime_officers);
```

7. List the criminal names for all criminals who have a less-than-average number of crimes and aren't listed as violent offenders.

8. List appeal information for each appeal that has a less than average number of days

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```
SELECT *  
FROM appeals  
WHERE AVG((filing_date - hearing_date)) < ALL (SELECT AVG((filing_date -  
hearing_date));
```

9. Create an alphabetical list of all criminals, including criminal ID, name, violent offender status, parole status, and any known aliases, by running and selecting which SQL statements will produce the correct output:

- a.

```
SELECT c.criminal_ID, c.last, c.firtdt, c.v_status, a.alias  
FROM criminals c, OUTER JOIN aliases a  
ON c.criminal_ID = a.criminal_ID;
```
- b.

```
SELECT c.criminal_ID, c.last, c.firtdt, c.v_status, c.p_status, a.alias  
FROM criminals c, LEFT OUTER JOIN aliases a  
ON c.criminal_ID = a.criminal_ID;
```
- c.

```
SELECT c.criminal_ID, c.last, c.firtdt, c.v_status, c.p_status, a.alias  
FROM criminals c, OUTER JOIN aliases a  
ON c.criminal_ID = a.criminal_ID;
```
- d.

```
SELECT c.criminal_ID, c.last, c.firtdt, c.v_status, c.p_status, a.alias  
FROM criminals c, NATURAL JOIN aliases a  
ON c.criminal_ID = a.criminal_ID;
```

10. List all attributes (*) from the following tables:

- a) Criminals, crimes, aliases, sentences, prob_officers,.officers,
crime_codes, appeals, crime_officers, crime_charges, appeals_id_sq,
prob_contact, criminal_dw